

Shadman Sakib

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EXPERIENCE

Missouri State University - Springfield, MO

Aug 2023 - Present

Graduate Research and Teaching Assistant

- Developed and deployed a reinforcement learning-driven cloud scheduling system, integrating **CloudSim** with **PyTorch DQN** models via **Flask API**, achieving 50%+ reduction in response time.
- Instructed Data Structures and Algorithms lab for **60+** students, designing curriculum and leading sessions, contributing to a **25% increase in average assignment scores**.

Girmairi - Remote (US based startup)

Sept 2021 - Feb 2025

Software Engineer

- Took ownership of architecting and scaling a full-stack dealership management platform using **React.js**, **Node.js**, **Express**, and **MongoDB**, enabling AI-assisted inventory workflows and cross-platform synchronization resulting in a streamlined process now employed by over 18 dealership locations, improving overall productivity.
- Led backend API development for web and mobile platforms, improving cross-platform data synchronization workflows and reducing average API response latency by **~40%**.
- Led the end-to-end development of the company's React Native mobile application, implementing scalable mobile infrastructure that supported **500+ active users** and maintained a **4.7-star rating** across the App Store and Google Play.
- Owned cloud infrastructure across development, staging, and production environments using **AWS EC2, S3, CloudFront, Route53, Elastic Beanstalk, WAF, and TLS**, sustaining **99.9% uptime** for client-facing systems.
- Spearheaded CI/CD automation using **AWS CodePipeline and CodeBuild**, reducing deployment cycles from manual releases to **~15-20 minutes** and accelerating production bug fixes and feature releases.

PROJECTS

- Conversational AI Chat Application:** Engineered a full-stack conversational AI platform with persistent chat memory using **Vue.js**, **Pinia**, **Node.js**, **Express**, **TypeScript**, and **PostgreSQL**, implementing real-time streaming responses and supporting **500+ concurrent chat sessions**.
- Real-Time Gesture-Based Navigation Interface:** Built a real-time computer vision interface using **OpenCV** and **MediaPipe**, achieving **94% gesture classification accuracy** while maintaining **30fps processing** for interactive navigation workflows.

EDUCATION

Missouri State University - Springfield, MO

Aug 2023 - Dec 2025

Master of Science in Computer Science | Data Science Certification | CGPA: 3.80/4.00

BRAC University - Dhaka, Bangladesh

Jan 2017 - Dec 2020

Bachelor of Science in Computer Science & Engineering | CGPA: 3.82/4.00

SKILLS

Languages: JavaScript (ES6+), TypeScript, Python, Java, C++, SQL, HTML5, CSS3, Bash/Shell

Frontend: React.js, Next.js, Vue.js, React Native, TailwindCSS, shadcn/ui, Pinia, Redux, Expo, Webpack

Backend: Node.js, Express.js, SpringBoot, Django, Flask, REST APIs, GraphQL, WebSockets

Databases: PostgreSQL, MySQL, MongoDB, Redis, Neon, Prisma ORM, Mongoose, Database Design, Query Optimization

Cloud & DevOps: AWS (EC2, S3, CloudFront, CodePipeline, CodeBuild, Route 53, SNS, CloudWatch, WAF, Elastic Beanstalk),

Docker, Jenkins, CI/CD, Nginx

AI / ML / Data: TensorFlow, PyTorch, Scikit-learn, Pandas, NumPy, Matplotlib, Seaborn, PowerBI, OpenCV

Tools & Practices: Git, GitHub, Jira, Figma, Agile/Scrum, System Design, AI-Assisted Development (Cursor, Claude Code)

PUBLICATIONS

- Reinforcement Learning Based Adaptive Task Scheduling in Cloud Environments, IEEE ICCT 2025
- Dynamic Load Balancing in Cloud Infrastructure for Energy Efficiency, IEEE Cloud Summit 2025
- ML & NLP-Based Mental Illness Classification, Springer ICICTD 2022